



Final Excel Project

IMDB Data Analysis

Project Objective

Comprehensive analysis of IMDB movie database
to support investor decision-making in the film industry

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1 Introduction

This Excel data analysis project focuses on the **Internet Movie Database (IMDB)** to develop actionable insights for film industry investment decisions. You will analyze comprehensive movie datasets using Excel's advanced functions, statistical methods, and visualization techniques.

1.1 Project Objectives

- Master Excel functions including VLOOKUP, statistical calculations, and data transformations
- Apply descriptive statistics and correlation analysis to identify trends
- Create effective visualizations and charts
- Generate a professional report with data-driven investment recommendations
- Present findings in a clear, compelling 10-15 minute presentation

2 Case Study: Investor's Dilemma

Scenario: You are assisting an investor interested in the movie industry but uncertain about where to allocate capital. Your role is to analyze IMDB data and provide insights that will guide strategic investment decisions.



Figure 1: Investor Dashboard Interface

2.1 Investor Profile

The investor has the following constraints and preferences:

Constraint	Description
Budget	Limited budget; focus on mid-range productions with strong ROI
Genre	Prefers Action, Drama, Comedy; open to other genres with proven profitability
Engagement	Values social media metrics (Facebook likes) as success indicators
Risk Tolerance	Risk-averse; requires data-driven insights to minimize uncertainty

Table 1: Investor Constraints and Preferences

2.2 Analysis Directions

Your analysis should explore one or more of these directions:

1. **Profitability by Genre** – Identify which genres yield highest profit margins and revenues

2. **Impact of Ratings** – Analyze correlation between movie ratings (G, PG, PG-13, R) and financial success
3. **Star Power Effect** – Examine relationship between actor popularity and gross revenue
4. **Budget vs. Profit** – Determine optimal budget ranges for maximum returns

3 Key Performance Indicators (KPIs)

3.1 Understanding KPIs

A **Key Performance Indicator (KPI)** is a quantifiable metric used to evaluate success in achieving specific objectives. In data analysis, KPIs transform raw data into actionable insights by focusing on critical metrics that drive success.

Important: Before selecting and using KPIs, please read the KPI guide available at:

<https://www.dropbox.com/scl/fi/y3a4h44cjp28yns2e60jd/KPIs.pdf?rlkey=7j8cq4cgyp14mpbjh3vi82iwh&dl=0>

3.2 Movie Industry KPI Examples

KPI	Formula	Purpose
Profit Margin	$\frac{\text{Revenue} - \text{Budget}}{\text{Revenue}}$	Financial efficiency
ROI	$\frac{\text{Revenue} - \text{Budget}}{\text{Budget}} \times 100\%$	Investment efficiency
Revenue per Minute	$\frac{\text{Revenue}}{\text{Duration}}$	Runtime effectiveness
Audience Engagement	Total Facebook Likes	Social media popularity

Table 2: Key Performance Indicators for Analysis

3.3 Your Task

1. Select a KPI relevant to your analysis goals
2. Justify why this KPI aligns with investor objectives
3. Calculate the KPI for each movie in the dataset
4. Identify trends and conditions where the KPI is maximized
5. Present findings in your report and presentation

4 Part I: Exploratory Data Analysis

4.1 Dataset Description

The IMDB dataset contains the following variables:

Variable	Description
<i>Categorical Variables</i>	
Title	Name of the movie
Release Date	Date the movie was released
Color/B&W	Whether filmed in color or black and white
Genre	Movie genre (Action, Comedy, Drama, etc.)
Language	Primary language spoken in the movie
Country	Country where the movie was produced
Rating	MPAA rating (G, PG, PG-13, R, etc.)
Lead Actor	Name of the lead actor
Director Name	Name of the director
<i>Quantitative Variables</i>	
Lead Actor FB Likes	Number of Facebook likes for the lead actor
Cast FB Likes	Total Facebook likes for the entire cast
Director FB Likes	Number of Facebook likes for the director
Movie FB Likes	Number of Facebook likes for the movie
IMDb Score	Rating on IMDB (scale 1-10)
Total Reviews	Total number of reviews on IMDB
Duration (min)	Movie length in minutes
Gross Revenue	Total revenue generated by the movie
Budget	Total production budget

Table 3: Dataset Variables

4.2 Required Tasks

4.2.1 Task 1: Data Integration

Employ **VLOOKUP** to merge the two tables (Sheet1 and Sheet2) into a unified dataset.

4.2.2 Task 2: Create New Variables

Generate the following new variables:

- **is_english:** Binary indicator (1 = English, 0 = Other languages)
- **Profit:** Gross Revenue - Budget
- **%Profit:** $\frac{\text{Gross Revenue}}{\text{Budget}} - 1$
- **duration_class:** Categorical ranges (e.g., "90-110 minutes" for a 100-min movie)

4.2.3 Task 3: Classify Variable Types

Specify the type for all columns:

Categorical (Genre, Rating) — **Binary** (Color/B&W, is_english) — **Discrete** (Total Reviews) — **Continuous** (Budget, Revenue)

4.2.4 Task 4: Descriptive Statistics

Calculate the following statistics for quantitative features in relation to categorical features (e.g., mean budget by rating, median revenue by genre):

Sum	Count	Average	Minimum
Maximum	10th Percentile	25th Percentile (Q1)	Median
75th Percentile (Q3)	90th Percentile	Standard Deviation	

Table 4: Descriptive Statistics to Calculate

4.2.5 Task 5: Visualization and Analysis

Create visualizations:

- Histograms (distribution analysis)
- Box plots (outlier identification)
- Bar charts (categorical comparisons)
- Scatter plots (relationship exploration)

Explore relationships: Genre vs. Budget, IMDb Score vs. Revenue, Actor Popularity vs. Success, Budget vs. Profit Margin

Ensure all charts include informative titles, axis labels, and legends.

5 Part II: Correlation Analysis

5.1 Understanding Correlation

Correlation analysis quantifies relationships between variables. The correlation coefficient (r) ranges from -1 to +1:

$r = +1$ Perfect positive	$r = 0$ No correlation	$r = -1$ Perfect negative
<i>Strength Interpretation</i>		
0.00-0.19: Very weak	0.40-0.59: Moderate	0.80-1.00: Very strong
0.20-0.39: Weak	0.60-0.79: Strong	

Table 5: Correlation Coefficient Interpretation

5.2 Application Examples

Example 1: IMDb Score and Revenue

Calculate correlation to determine if higher-rated movies generate more revenue. A positive correlation (e.g., $r = 0.7$) supports the conclusion that quality drives financial success. Use scatter plots with trend lines to visualize.

Example 2: Lead Actor Popularity and Revenue

Examine correlation between actor Facebook likes and gross revenue. Strong positive correlation suggests investing in movies with popular actors leads to higher financial gains.

5.3 Best Practices

- Use Excel's CORREL function to calculate coefficients
- Create scatter plots with trend lines to visualize relationships
- Interpret results in context of business objectives
- Remember: correlation does not imply causation

6 Part III: Example Insights

6.1 Sample Findings

These hypothetical examples guide your analytical approach:

6.1.1 Genre Profitability

Finding: Action genre movies show highest average profit margins (45%) with consistent performance across budget ranges.

Investment Implication: Safer bet for financial returns with strong international appeal.

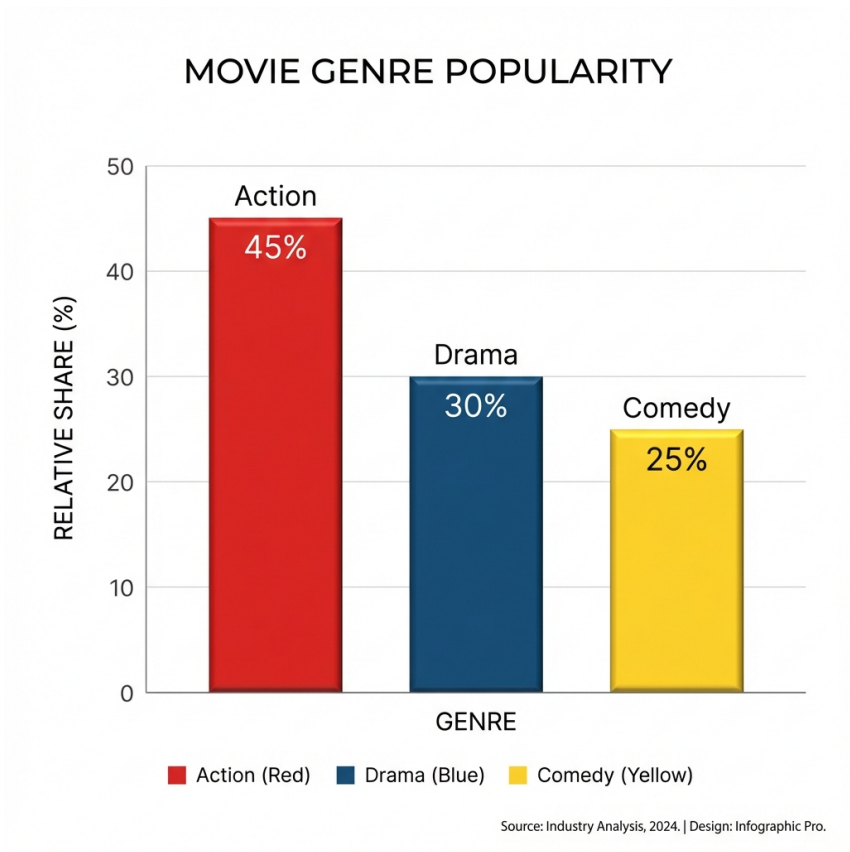


Figure 2: Comparative Profitability by Genre

6.1.2 Star Power Effect

Finding: Films with lead actors having 1M+ Facebook likes show 25% higher revenue ($r = 0.65$ correlation).

Investment Implication: Prioritize projects with established stars for pre-built audience anticipation.

6.2 Presentation Guidelines

When presenting your findings:

- Support conclusions with visualizations and specific numbers
- Connect findings directly to investor objectives
- Acknowledge data limitations and assumptions
- You may use external sources (industry reports, academic research) to strengthen your analysis and claims

7 Part IV: Presentation of Findings

7.1 Presentation Structure

Your **7-minute recorded video presentation** should include:

1. **Objective** – Analysis goals and investor context
2. **Data Summary** – Dataset overview and key characteristics
3. **Methodology** – Excel functions and analytical approaches used
4. **Key Findings** – Answers to research questions with supporting visuals
5. **KPI Analysis** – Selected KPI, calculations, and maximization insights
6. **Recommendations** – Specific, actionable investment advice
7. **Q&A** – Prepared responses to anticipated questions

7.2 Recording Requirements

Video Submission Guidelines:

- **Duration:** Exactly 7 minutes (± 30 seconds acceptable)
- **Format:** MP4, AVI, or MOV file format
- **Content:** Screen recording of your Excel analysis with voice narration
- **Quality:** Clear audio, visible charts and data
- **Submission:** Upload to designated platform and send link to instructor

Example Presentation:

For reference, you may review a sample student presentation (note: there is no single correct solution):

<https://www.dropbox.com/scl/fi/zuxknhq792zes6gawx89w/Final-Project.pptx?rlkey=zz65hdg6lg0ksbeex1owq44vg&dl=0>

7.3 Presentation Best Practices

Aspect	Guidelines
Clarity	One key message per slide; avoid information overload
Visual Appeal	Consistent design; readable fonts; high-quality charts
Storytelling	Logical flow; build to compelling conclusions
Data Visualization	Clear labels, appropriate chart types, highlight key findings
Time Management	Practice to stay within 7-minute limit; prioritize key insights
Audio Quality	Clear narration; minimize background noise; speak confidently

Table 6: Presentation Guidelines

8 Final Remarks

8.1 Success Factors

To excel in this project:

- **Be thorough** – Complete tasks with attention to detail
- **Think critically** – Interpret results, don't just calculate
- **Stay focused** – Connect all analysis to investor objectives
- **Communicate clearly** – Make complex data accessible
- **Prioritize quality** – A focused, well-executed analysis is more valuable than superficial coverage

8.2 Important Notes

You are **not required** to complete all tasks unless they are relevant to your specific analysis goals. Focus on tasks that align with your objectives and support your insights effectively.

Quality over quantity – choose depth over breadth.

Using External Sources:

You are **encouraged to use external sources** (industry reports, academic papers, market research) to strengthen your analysis and support your claims. Proper citation of sources adds credibility to your findings.

8.3 Skills Developed

This project will demonstrate your proficiency in:

- Advanced Excel functions and statistical analysis
- Data visualization and chart creation
- Business insight development and communication
- Professional presentation and recording skills

These skills are directly applicable to data analyst, business intelligence, financial analysis, and consulting roles.

Good luck with your analysis!